

Sales Forecasting & Demand Intelligence System

Executive Business Report — Prepared by Pavani | Internship Week 3 & 4 | July 2026

Executive Summary

This report presents findings from an intelligent sales forecasting system built on four years of Superstore retail data covering 9,994 transactions across all product categories and regions. Using three different forecasting approaches, the system predicts the next three months of sales with measurable accuracy, identifies weeks of unusually high or low sales, and groups products into four distinct demand categories — each requiring a different stocking strategy. The goal is to help the supply chain team make smarter inventory decisions before problems occur, not after stock runs out or piles up in the warehouse.

Key Findings from Data Analysis

Revenue breakdown: Technology is the highest-earning category at approximately \$836,000 total over 4 years, followed by Furniture and Office Supplies. However, Office Supplies has the highest number of individual orders — meaning it drives volume while Technology drives value.

Regional performance: The West region shows the most consistent year-over-year growth and should be the primary focus for capacity planning and restocking priority.

Seasonality: November and December consistently generate 35-45% higher sales than the annual monthly average — every single year. March shows a secondary spike. January is consistently the weakest month.

Shipping time: Average time from order to shipment is 3.9 days. Central region is slightly slower at 4.1 days on average — this may be worth reviewing with logistics.

3-Month Sales Forecast

Three forecasting models were built and compared: a statistical model (SARIMA), Facebook's Prophet (an industry-standard tool), and a machine learning model (XGBoost). Prophet performed best overall and is recommended for ongoing use.

Month	Forecasted Sales	Expected Range
Month 1	\$58,000 – \$62,000	\$52,000 – \$68,000
Month 2	\$61,000 – \$65,000	\$54,000 – \$72,000
Month 3	\$67,000 – \$73,000	\$59,000 – \$81,000

Sales are on a growth trajectory. Month 3 shows a noticeable step up — consistent with the historical pattern of stronger Q4 demand. The supply chain team should begin building buffer stock by the end of Month 2.

Top 3 Sales Anomalies Detected

1. Late November spikes (all 4 years): Sales jump 40-60% above normal in the final week of November. Likely cause: year-end corporate purchasing and festive promotions.
Recommendation: pre-position stock 3 weeks in advance.

- 2. September mid-month surge:** A consistent secondary spike appears in the second week of September each year. Likely cause: back-to-school and new fiscal quarter procurement by business customers buying office supplies and technology.
- 3. January drop:** Sales fall 25-35% below the annual average every January without exception. Likely cause: post-holiday budget freezes and reduced consumer spending. Overstocking going into January should be avoided.

Product Demand Segmentation & Stocking Strategy

Segment	Sub-Categories	Recommended Action
High Volume, Stable	Binders, Paper, Storage	Automate reordering. Keep 4-week safety stock.
Growing Demand	Phones, Chairs, Copiers	Increase orders proactively. Negotiate bulk pricing.
Low Volume, High Volatility	Bookcases, Supplies	Keep lean stock. Maintain fast supplier access.
Declining / Niche	Labels, Envelopes, Fasteners	Reduce stock levels. Run promotions to clear inventory.

3 Business Recommendations

- 1. Build a November pre-stocking protocol.** Data shows November spikes every year without exception. Begin increasing inventory for Technology and Office Supplies by October 15th. A 40% buffer above the October average is a reasonable starting point based on historical patterns.
- 2. Prioritise West region for capacity expansion.** The West consistently leads in both total sales and growth rate. Any new warehouse investment or logistics partnership should be West-first to serve the highest-demand geography.
- 3. Review Declining segment products quarterly.** Fasteners, Labels, and Envelopes show declining or flat demand with low average order values. Maintaining standard stock levels for these items ties up working capital unnecessarily. A quarterly review and targeted clearance promotion would free up an estimated 8-12% of storage space.

System Limitation

This forecasting system is trained on historical patterns. It cannot account for sudden external disruptions — a supplier shutdown, a new competitor entering the market, or a macroeconomic shock. Predictions should be treated as an informed starting point, not a guarantee. The model should be retrained every quarter with fresh sales data to remain accurate.